

Meri Task API — Aaj Ka Kaam (Recap Notes)

Ye file maine banai hai taake future mein yaad rahe ke maine kya kya kiya tha aur kyun. Agar dubara isi tarah ka kaam karna ho, to ye padh lena — sab yaad aa jayega.

1. Maqsad Kya Tha

Ek chhoti si **CRUD API** banani thi — matlab ek server jo to-do list (tasks) manage kar sake:

- **Create** — naya task banana
- **Read** — tasks dekhna
- **Update** — task ko badalna
- **Delete** — task ko hatana

2. Tools Jo Use Kiye

Tool	Kaam
Python	Programming language, jismein code likha
VS Code	Wo app jahan code likha aur chalaya
FastAPI	Python ka framework, jisse server bana
Uvicorn	Server ko chalane wala tool
Swagger UI	Ek visual page (/docs) jahan API ko browser se test kiya bina coding ke
Git & GitHub	Code ko online, public save karne ka tareeqa

3. Steps Jo Follow Kiye (Order Mein)

Step 1 — Setup

- Python install kiya
- VS Code install kiya
- Ek folder banaya: `f1yrank_crudapi`
- Us folder ke andar ek **venv** (virtual environment) banaya — isse Python ke tools sirf isi project ke liye alag rehte hain
- `pip install fastapi uvicorn` se zaroori tools install kiye

Step 2 — Server Shuru Kiya (Stage 0)

- `main.py` naam ki file banayi
- Sabse simple code likha: server "Hello" bole jab koi usse baat kare
- Terminal mein `uvicorn main:app --reload` likh kar server chalaya
- Browser mein `http://localhost:8000` khol kar check kiya

Step 3 — Endpoints Banaye (Stage 1-4)

Ek ek karke ye "doors" (endpoints) code mein add kiye:

Method	Endpoint	Kaam
GET	/	API ka intro
GET	/health	Server zinda hai check karna
GET	/tasks	Saari tasks dikhana
GET	/tasks/{id}	Ek specific task dikhana
POST	/tasks	Naya task banana
PUT	/tasks/{id}	Task ko update karna
DELETE	/tasks/{id}	Task ko delete karna

- Har naye endpoint ke baad file **save** ki (Ctrl+S), server khud restart ho jata tha
- Har baar browser ya terminal se test kiya ke sahi kaam kar raha hai ya nahi

Step 4 — Testing Ke Tareeqe

- **Browser** — sirf GET endpoints ke liye (address bar mein URL likh kar)
- **PowerShell (Invoke-RestMethod)** — POST, PUT, DELETE test karne ke liye, kyunke browser se ye nahi hota
- **Swagger UI (/docs)** — sabse aasan tareeqa, bina code likhe, buttons dabakar test karna

Step 5 — Swagger UI Se Poora Test Kiya

- `http://localhost:8000/docs` khola
- Har endpoint ko "Try it out" → "Execute" dabakar test kiya
- Screenshots liye

Step 6 — GitHub Par Publish Kiya

- GitHub par account tha, ek **public repository** banayi: `flyrank_first-assignment-`
- Git install kiya (computer mein pehle se nahi tha)
- `README.md` file banayi — jismein likha: ye project kya hai, kaise chalayein, endpoints ki list, ek test ka example
- `.gitignore` file banayi — taake `venv` folder GitHub par na jaye
- Git commands se code ko GitHub par push kiya:
 - `git init` → Git shuru karna
 - `git add .` → files ko taiyar karna
 - `git commit -m "message"` → ek "save point" banana
 - `git push origin main` → GitHub par bhejna

4. Zaroori Cheezein Jo Yaad Rakhni Hain

1. **Server hamesha localhost par chalta hai** — matlab sirf apne computer tak mehdood, koi bahar wala access nahi kar sakta jab tak GitHub par push na karein
2. **GitHub par sirf code jata hai** (jo humne khud likha), koi personal file ya password nahi jata
3. **venv folder kabhi GitHub par nahi jana chahiye** — isliye `.gitignore` zaroori hai
4. **PowerShell mein POST/PUT/DELETE ke liye Invoke-RestMethod use karna** — `curl.exe` PowerShell mein thoda tricky hota hai
5. **Git commits** = ek "save note" jisme likha ho "maine ye kiya" — jitni zyada commits, utna zyada pata chalta hai kaam kaise hua

5. Agar Dubara Karna Ho To Kahan Se Shuru Karein



bash

```
cd Desktop\flyrank_crudapi
venv\Scripts\activate
uvicorn main:app --reload
```

Phir browser mein: `http://localhost:8000/docs`

6. Mera GitHub Repo Ka Link



`https://github.com/lhub2856-bit/flyrank_first-assignment-.git`

Ye notes maine khud apne liye banai hain taake agli baar bhoolun nahi.